

An Empirical Comparison of Human-Elicited and LLM-Generated Functional Requirements in a Physical Therapy Follow-Up System: A Paired-Analyst Study in Ecuador

Kevin Germán Contreras Chávez

Angelo Paul Zambrano Moya

Universidad Técnica Estatal de Quevedo (UTEQ), Ecuador

Software Engineering Program – Requirements Engineering (ISR-401)

Corresponding author: Angelo Paul Zambrano Moya

ORCID: 0009-0006-0056-8482

Manuscript version 2.0 – September 2026

Target journal: *Requirements Engineering* (Springer Nature)

Abstract

Digital support for home-based physical therapy depends on requirements that faithfully capture the needs of patients, caregivers, and physiotherapists. Large Language Models (LLMs) are increasingly proposed as assistants for requirements elicitation, but their comparative quality against human-elicited requirements remains underexplored in real-world, resource-constrained software engineering education settings. This paper reports a paired-analyst empirical study comparing the quality of functional requirements (FR) elicited by a human requirements engineering process against FR generated by an LLM (GPT-5.5-mini) from the same 18 anonymized interview transcripts, for the Intelligent System for Physical Therapy Control and Follow-up (SICST). The protocol, hypotheses, and analysis plan were preregistered on the Open Science Framework (OSF, DOI 10.17605/OSF.IO/82Q76) prior to data generation. Thirty-three human-elicited and thirty-three LLM-generated FR (66 total) were anonymized and blind-rated by four independent judges across five quality dimensions (completeness, absence of ambiguity, verifiability, source correctness, internal consistency) on a 1–5 Likert scale. Inter-rater agreement was quantified with Fleiss' kappa. Eleven requirement pairs with strict thematic correspondence were compared using paired *t*-tests or Wilcoxon signed-rank tests, with Holm–Bonferroni correction for multiple comparisons. Inter-rater agreement was low to negligible across dimensions (Fleiss' κ ranging from -0.011 to 0.167). No dimension showed a statistically significant difference between human- and LLM-generated requirements after correction (all adjusted $p > 0.05$). These findings suggest that, in this small-sample paired comparison, LLM-generated functional requirements were not distinguishable in quality from human-elicited ones, while highlighting the need for larger samples and more reliable evaluation protocols in future replications.

Keywords: Requirements elicitation; Physiotherapy informatics; Large Language Models; Empirical study; Patient tracking.

1 Introduction

Physical therapy frequently extends beyond the clinical session. Patients may be expected to remember exercises, execute movements correctly, respect repetition and rest instructions, report pain or fatigue, and communicate problems between appointments. These activities create a requirements problem before they create an algorithmic problem: a digital system must first represent the needs, constraints, responsibilities, and safety expectations of patients, caregivers, and professionals in a way that can be verified and traced. Understanding whether Large Language Models can support this elicitation step, and at what quality cost, matters directly to small teams that lack dedicated requirements analysts.

Requirements engineering (RE) provides the methods needed to elicit, analyze, document, validate, prioritize, and manage stakeholder expectations (Pohl, 2010; ISO/IEC/IEEE, 2018; Sommerville, 2016). Recent work has begun to examine whether LLMs can support or replace parts of this process, reporting mixed results on completeness, ambiguity, and traceability when LLM output is compared against human analysts (Cheng et al., 2026). However, most of this evidence comes from generic or large-scale software domains; comparative studies in small, real-world, health-adjacent domains such as home-based physical therapy remain scarce, and few report a preregistered, blind, paired evaluation design.

SICST (*Sistema Inteligente de Control y Seguimiento de Terapia Física*) was developed as an academic case study in Requirements Engineering at Universidad Técnica Estatal de Quevedo. The project addresses follow-up of physical therapy routines by separating patient and physiotherapist interactions while maintaining a common traceability chain from field evidence to requirements, design elements, and prototype behavior. This paper reports the terminal empirical component of that project: a controlled comparison of requirements quality between the team’s own elicitation process and an LLM prompted with the same underlying field evidence.

This study contributes: (1) a preregistered, blind, paired-analyst protocol for comparing human- and LLM-generated functional requirements in a real, non-simulated domain; (2) an empirical estimate of the quality gap (or lack thereof) between both sources across five standard RE quality dimensions; and (3) a transparent report of inter-rater reliability challenges when four independent, non-expert judges rate requirements quality on a Likert scale, which we discuss as a methodological caution for similar undergraduate empirical RE studies.

RQ1. Do quality dimensions (completeness, absence of ambiguity, verifiability, source correctness, internal consistency) differ between functional requirements elicited by a human requirements engineering process and those generated by a Large Language Model, when both are derived from the same anonymized interview transcriptions?

H0. There are no statistically significant differences in the quality of functional requirements generated through a human process versus those generated through a language model.

H1. There are statistically significant differences in the quality of functional requirements generated through a human process versus those generated through a language model.

2 Background and Related Work

2.1 LLMs for requirements elicitation

Generative language models have been proposed for several RE sub-tasks, including requirements extraction from unstructured text, ambiguity detection, and requirements quality assessment (Cheng et al., 2026). Evidence on elicitation quality specifically is more limited: most published comparisons use public or synthetic corpora rather than transcripts from a live, IRB-approved field study, and few preregister the comparison protocol before generation, which leaves open the possibility of post-hoc rationalization of favorable results.

2.2 Requirements quality dimensions

Completeness, absence of ambiguity, verifiability, traceability to source, and internal consistency are long-standing quality attributes for individual requirements statements (ISO/IEC/IEEE, 2018; Sommerville, 2016). We adopt five of these dimensions, adapted to allow blind comparison between two requirement sources drawn from the same evidentiary base, rather than an absolute compliance checklist.

2.3 Physical therapy follow-up and telerehabilitation

Digital follow-up of physical therapy raises additional constraints beyond generic software requirements: exercise adherence, pain and fatigue self-reporting, caregiver involvement, and the professional’s authority to override any automated suggestion (ISO/IEC, 2023; Chazette and Schneider, 2020). These constraints motivated the original field study from which both requirement sets (human and LLM) were derived.

2.4 Inter-rater agreement in requirements evaluation

Fleiss' kappa is a standard measure of agreement among three or more raters using categorical or ordinal scales (Fleiss, 1971; Cohen, 1960). Low agreement in subjective quality ratings is a known risk in small-panel, non-expert evaluation designs and is reported transparently here rather than adjusted away.

3 Methodology

3.1 Design

This is a non-experimental, paired comparative study. No causal relationship between requirement source and quality is inferred; the study only compares the perceived quality of two requirement sets drawn from the same source evidence. The protocol, hypotheses, blinding procedure, and analysis plan were registered on OSF (DOI 10.17605/OSF.IO/82Q76) on 2026-09-12, before the LLM-generated requirements or the blind evaluation data existed.

3.2 Population and sampling

The unit of analysis is the individual functional requirement (FR). Two comparison groups were constructed from the same evidentiary source:

- **Human-elicited group** ($n = 33$): FR produced through the team's standard requirements engineering process (interviews, thematic analysis, and consolidation into the project's ERS/SRS).
- **LLM-generated group** ($n = 33$): FR produced by prompting GPT-5.5-mini with the same 18 anonymized interview transcripts used for human elicitation (physiotherapists, family caregivers/patients, and validation walkthrough participants), under a strict instruction set requiring each requirement to be grounded exclusively in the supplied transcripts (see Section 3.3).

Both sets were combined into a single list of 66 anonymized items and presented to four independent, blind judges (Section 3.4).

3.3 LLM requirements generation

The LLM (GPT-5.5-mini, accessed via ChatGPT) was given, in a single turn, the full text of the 18 anonymized interview transcripts and an explicit rule set requiring: (a) grounding every requirement strictly in the supplied transcripts, with no external knowledge or invented functionality; (b) a fixed sentence template ("*The system shall [verifiable action] [object/condition]*"); (c) specificity, verifiability, and traceability to a concrete transcript fragment; (d) no duplicate or overlapping requirements; (e) sequential numbering (RF-01 to RF-33); and (f) output restricted to a CSV with columns `id_requisito`, `descripcion`, with no additional prose. A second turn asked the model to confirm its exact version and confirm that no external information was used. The full prompt, both turns, the model's confirmation, and the resulting CSV are versioned in `06_Experimento/prompts_llm/` of the project repository.

3.4 Blinding and evaluation instrument

The 66 FR (33 human, 33 LLM) were normalized to a common format, shuffled, and assigned anonymous item identifiers with no indication of source. Four independent judges, blind to the origin of each item, rated every item on a 1–5 Likert scale across five dimensions: completeness, absence of ambiguity, verifiability, source correctness, and internal consistency. The private key mapping each anonymized item back to its real requirement identifier and source (human/LLM) was stored encrypted and was not consulted until after all four evaluations were collected.

3.5 Thematic pairing

To support a paired statistical comparison, human and LLM requirements addressing the same underlying interview theme were matched by two independent reviewers into a traceability matrix (`matriz trazabilidad tema_RF.csv`).

Eleven pairs achieved strict, unambiguous thematic correspondence (type **pareado**); these are the pairs used in the confirmatory paired analysis reported in Section 4. The remaining items (partial matches, weak matches, or items with no thematic counterpart in the other set) are retained in the traceability matrix for transparency but excluded from the confirmatory test, since they do not support a valid 1-to-1 paired comparison.

3.6 Statistical analysis plan

For each of the five quality dimensions and for the aggregate score, the analysis (i) computed Fleiss’ kappa across all four raters and 66 items to quantify inter-rater agreement; (ii) tested normality of the 11 within-pair differences (human – LLM) with the Shapiro–Wilk test; (iii) applied a paired t -test when normality was not rejected, or a Wilcoxon signed-rank test otherwise; and (iv) corrected the five dimension-level p -values for multiple comparisons using the Holm–Bonferroni procedure. A 95% bootstrap confidence interval (10,000 resamples) was computed for the mean paired difference. All analyses were executed by a single versioned script (`06_Experimento/scripts_analisis/analisis_experimento_llm_humano.py`) that reproduces every table and figure reported below directly from the raw blind-rating spreadsheets.

3.7 Deviations from the preregistered protocol

The exact statistical test per dimension (paired t vs. Wilcoxon) and the Holm–Bonferroni correction were specified operationally after registration, since the OSF protocol described the general comparison strategy without fixing the individual test names; this operational rule was fixed *before* the data were inspected for significance, and is documented in `06_Experimento/osf_deviations.pdf`. No other deviation from the registered design occurred.

4 Results

4.1 Inter-rater agreement

Agreement among the four blind raters was low to negligible across all five dimensions (Table 1). This indicates that the raters frequently disagreed on how to score individual requirements on the 1–5 scale, independently of requirement source.

Table 1: Fleiss’ kappa by quality dimension (all 66 items, 4 raters).

Dimension	Fleiss’ κ
Completeness	0.132
Absence of ambiguity	0.167
Verifiability	−0.013
Source correctness	−0.011
Internal consistency	0.046

4.2 Paired comparison (RQ1)

Table 2 reports the paired analysis over the 11 thematically-matched requirement pairs. All five within-pair difference distributions did not reject normality (Shapiro–Wilk $p > 0.05$), so a paired t -test was used throughout. After Holm–Bonferroni correction for the five dimension-level comparisons, no dimension reached statistical significance.

Table 2: Paired comparison, human-elicited vs. LLM-generated FR ($n = 11$ pairs). Mean difference is human minus LLM.

Dimension	$t(10)$	Mean diff.	95% CI	p (raw)	p (Holm)
Completeness	0.714	-0.136	[-0.341, 0.341]	0.492	1.000
Absence of ambiguity	-0.468	-0.091	[-0.477, 0.250]	0.650	1.000
Verifiability	1.000	0.136	[-0.136, 0.386]	0.341	1.000
Source correctness	1.550	0.159	[-0.045, 0.341]	0.152	0.761
Internal consistency	0.938	0.091	[-0.091, 0.273]	0.371	1.000

Figure 1 shows the paired global scores (average of the five dimensions) for each of the 11 matched themes, illustrating the substantial overlap between the two sources.

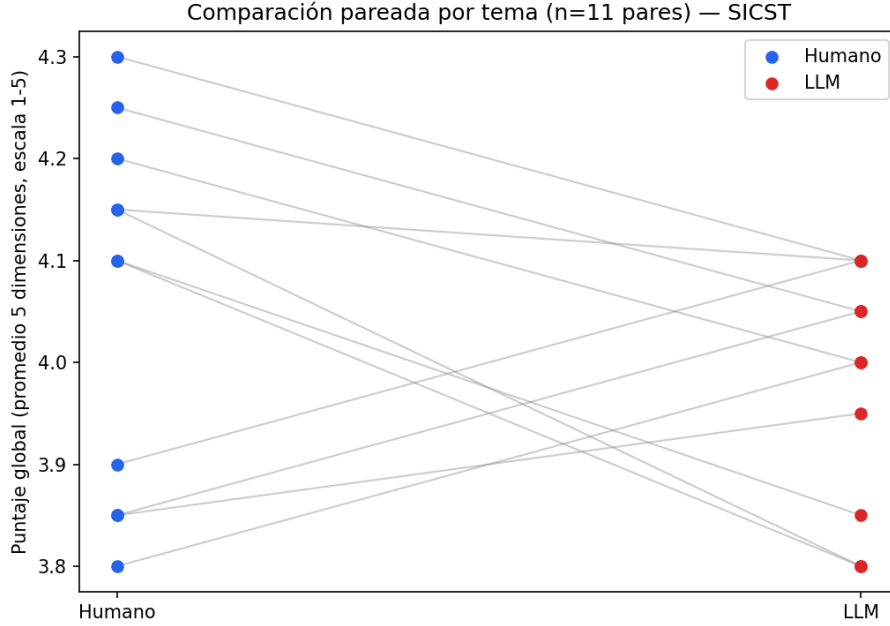


Figure 1: Paired global quality score by thematic pair, human-elicited vs. LLM-generated requirements ($n = 11$).

We cannot reject H_0 for any of the five quality dimensions. Within the limits of this small-sample design, human-elicited and LLM-generated functional requirements were not distinguishable in perceived quality.

5 Discussion

5.1 Implications for research

The absence of a detectable quality gap, combined with very low inter-rater agreement, is itself an informative result: it suggests that, for short, templated functional requirements derived from the same evidentiary transcripts, non-expert raters struggle to reliably distinguish machine-generated from human-generated statements. This is consistent with the broader concern that LLM output can closely mimic the surface form of expert-authored technical text (Cheng et al., 2026), which may complicate quality assurance processes that rely on manual review.

5.2 Implications for practice

For small teams without dedicated requirements specialists, an LLM prompted directly and strictly from anonymized field transcripts produced requirements that blind raters did not judge as worse, on average, than the team’s own elicitation process. This does not license a general recommendation to replace human elicitation; it does suggest that LLM-assisted drafting, followed by human review and thematic validation, is a defensible starting point for resource-constrained projects.

5.3 A surprising or counterintuitive finding

The negative and near-zero Fleiss’ kappa values for verifiability and source correctness are notable: they indicate that, on those two dimensions, the four raters agreed less than chance would predict. This is unusual and suggests the rating instrument itself, not only the requirements, deserves revision in future replications (e.g., anchored behavioral examples per Likert point, or a smaller, better-trained rater panel).

We restrict all conclusions to this specific, small-sample, Ecuadorian, physical-therapy domain study; no claim is made that LLMs match human analysts in general software domains, sample sizes, or languages.

6 Threats to Validity

Following common software-engineering case-study practice (Runeson and Höst, 2009; Wohlin et al., 2012), threats are organized into construct, internal, external, and conclusion validity.

6.1 Construct validity

The five quality dimensions operationalize a subjective judgment on a 1–5 scale without behaviorally anchored rating examples, which may partly explain the low inter-rater agreement observed. The confirmatory paired test uses only the 11 strictly-matched pairs; broader, weaker, or unmatched items (55 of 66) are not part of the confirmatory statistical claim.

6.2 Internal validity

Raters were blind to requirement origin, which mitigates expectation bias, but they were not trained requirements engineers, which may explain the low kappa values. The Holm–Bonferroni correction and the choice between paired t and Wilcoxon tests were fixed by a rule specified before inspecting results (Section 4 deviations), reducing the risk of HARKing or p -hacking, though the operational rule itself was specified after, not before, OSF registration.

6.3 External validity

The work concerns one academic project, one LLM (GPT-5.5-mini), one prompting strategy, and a physical-therapy domain in a specific Ecuadorian context. Results should not be generalized to other LLMs, prompting strategies, domains, or rater populations without further replication.

6.4 Conclusion validity

With only 11 paired observations, statistical power to detect small-to-moderate effect sizes is limited; the failure to reject H_0 should not be interpreted as strong evidence of true equivalence, but as an inconclusive result under low power, reported transparently rather than reframed as a positive finding.

7 Ethics, Privacy, and Data Governance

The underlying interview transcripts used to generate both requirement sets were collected under the project’s approved informed-consent process and are stored anonymized, with direct identifiers (names, national identification numbers, signatures) restricted to an encrypted, access-controlled container separate from the public repository. The blind evaluation data and the private de-anonymization key are likewise separated: the key is stored encrypted and was not consulted until after all four evaluations were collected, preserving both participant privacy and the integrity of the blinding procedure.

8 Reproducibility and Open Materials

The complete experimental package is versioned in the project repository under 06_Experimento/: the preregistered protocol (`protocolo.pdf`), the OSF registration receipt (`osf_registration.pdf`), the deviations declaration (`osf_deviations.pdf`), the exact LLM prompts and confirmations (`prompts_llm/`),

the raw blind-rating spreadsheets and the LLM-generated requirements (`datos_crudos/`), the thematic pairing matrix (`datos_procesados/`), the single reproducible analysis script (`scripts_analisis/`), and every table and figure reported in this paper (`resultados/`). Running the analysis script against the raw data reproduces Tables 1–2 and Figure 1 exactly.

Dataset DOI: <https://doi.org/10.5281/zenodo.22315298> | OSF registration:
<https://doi.org/10.17605/OSF.IO/82Q76>

9 Conclusion

This paper reported a preregistered, blind, paired-analyst comparison of the quality of functional requirements elicited by a human process versus generated by an LLM (GPT-5.5-mini), for a physical therapy follow-up system, using the same 18 anonymized interview transcripts as the evidentiary source for both. Inter-rater agreement among four blind judges was low to negligible across all five evaluated quality dimensions. After correcting for multiple comparisons, no dimension showed a statistically significant difference between human-elicited and LLM-generated requirements (all adjusted $p > 0.05$); the null hypothesis could not be rejected.

These results, obtained under a small paired sample ($n = 11$), should be read as a transparent, inconclusive finding rather than evidence that LLMs match human requirements engineers in general. Future work should increase the number of thematically-matched pairs, train and calibrate the rater panel to improve inter-rater agreement, and replicate the comparison across additional LLMs and domains before drawing practice-level conclusions.

Declarations

Funding. This academic project reports no external research funding in the materials used for this manuscript.

Conflict of interest. The authors declare no conflict of interest relevant to this manuscript.

Author contributions. Angelo Paul Zambrano Moya led the OSF preregistration, LLM requirements generation, blind-evaluation coordination, and statistical analysis. Kevin Germán Contreras Chávez contributed as a co-interviewer in field data collection sessions conducted remotely. Both authors reviewed the manuscript and are responsible for verifying that no evidence reported here was fabricated or altered.

Data availability. The reproducible dataset is available at Zenodo, DOI: <https://doi.org/10.5281/zenodo.22315298>. The preregistered protocol is available at OSF, DOI: <https://doi.org/10.17605/OSF.IO/82Q76>. Public materials exclude direct identifiers and signed consent documents.

Code availability. The project source repository, including the analysis script that reproduces every reported table and figure, is publicly available at https://github.com/AngeloP2114/Sistema_Terapia_Fisica_ISR401_2B.

Use of AI tools. An LLM (GPT-5.5-mini) was used to generate one of the two compared requirement sets, as the object of study itself; this use is fully documented in `06_Experimento/prompts_llm/` and is not a writing aid. Separately, AI assistance (Claude, Anthropic) was used to support repository organization, data-processing script development, and manuscript drafting from author-provided real data and results; no participant, response, or empirical result was invented or altered by this assistance.

References

- Chazette, L. and Schneider, K. (2020). Explainability as a non-functional requirement: challenges and recommendations. *Requirements Engineering*, 25:493–514.
- Cheng, H., Husen, J. H., Lu, Y., Racharak, T., Yoshioka, N., Ubayashi, N., and Washizaki, H. (2026). Generative AI for requirements engineering: A systematic literature review. *Software: Practice and Experience*.
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46.

- Fleiss, J. L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382.
- ISO/IEC (2023). ISO/IEC 25010:2023 Systems and Software Engineering – SQuaRE – Product Quality Model.
- ISO/IEC/IEEE (2018). ISO/IEC/IEEE 29148:2018 Systems and Software Engineering – Life Cycle Processes – Requirements Engineering.
- Pohl, K. (2010). *Requirements Engineering: Fundamentals, Principles, and Techniques*. Springer, Berlin, Germany.
- Runeson, P. and Höst, M. (2009). Guidelines for conducting and reporting case study research in software engineering. *Empirical Software Engineering*, 14:131–164.
- Sommerville, I. (2016). *Software Engineering*. Pearson, Boston, MA, USA, 10 edition.
- Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., and Wesslén, A. (2012). *Experimentation in Software Engineering*. Springer, Berlin, Germany.